doi:10.11772/j.issn.1001-9081.yyyymmnnnn

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NAME Name[1,2], NAME Name[1*], NAME Name[2]
(1.College of ****, University, City Province Zip Code, China;
2.College of ****, University, City Province Zip Code, China)

Abstract:
https://www.joca.cn/CN/column/column18.shtml

Keywords: keyword; keyword; keyword; keyword; keyword (TCP IP ;
Peer-to-Peer (P2P); Wireless Sensor Network (WSN))

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https://www.joca.cn/CN/column/column15.shtml
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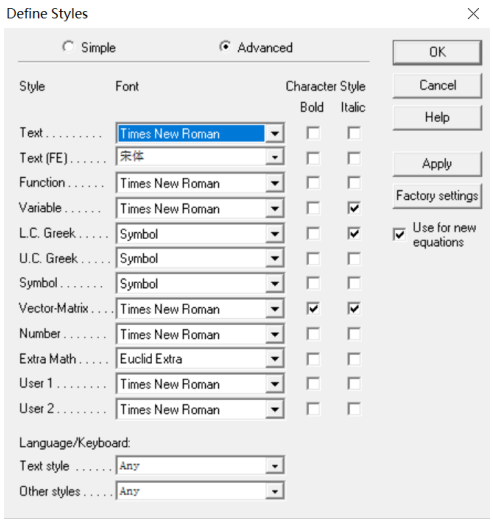
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This work is partially supported by the National High Technology Research and Development Program (863 Program) of China (12345678), the National Basic Research Program (973 Program) of China (87654321), the National Natural Science Foundation of China (22345678, 23456789), the Fundamental Research Funds for the Central University (20150906).

ZHANG Shanshan, born in 1967, Ph. D., professor. Her research interests include rough set, interval set, intelligent information processing, Web intelligence, data mining.
LI Si, born in 1990, M. S. candidate. His research interests include rough set, granular computing, machine learning, data mining, knowledge technology, cognitive computing.
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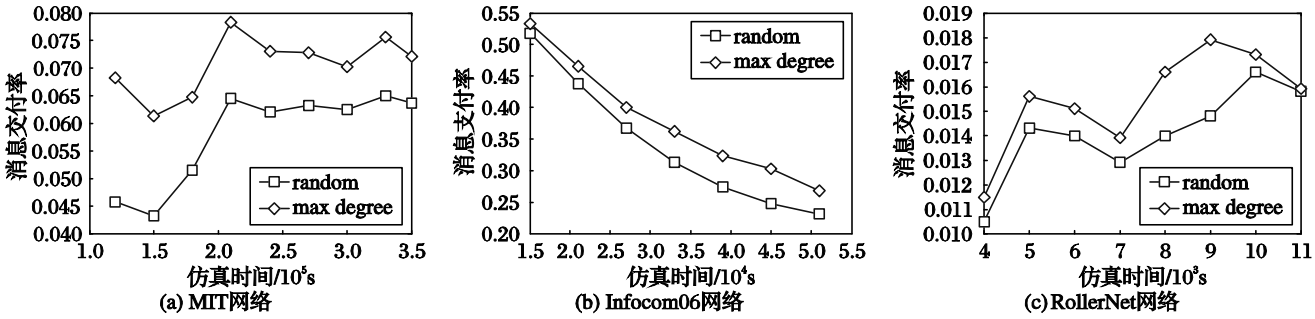
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Fig. 1 Comparison of message delivery rates of three network

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1 UCI

Tab. 1 UCI dataset used in the experiment

bupa	2	345	6
seeds	3	210	7
ecoli8	8	336	7
yeast	10	1484	8
vehicle	4	846	18
glass	7	214	39

2

Tab. 2 Comparison of single-sample recognition rate and network training time for each classification algorithm

	/%				/s
	10	100	300	466	
ELM	90	46.67	41.96	37.99	18.96
PNN	100	93.00	86.65	78.07	30.24
FA	100	93.33	91.29	87.15	24.88

5

$$Sim(C_1,C_2)=\sum_{i=1}^{4}\beta_i Sim_i(C_1,C_2) \tag{1}$$

$$\boldsymbol{A}=\begin{bmatrix} x^{11} & x^{12} & \cdots & x^{1n} \\ x^{21} & x^{22} & \cdots & x^{2n} \\ \vdots & \vdots & & \vdots \\ x^{m1} & x^{m2} & \cdots & x^{mn} \end{bmatrix} \tag{2}$$

$$\boldsymbol{v}_i=\sum_{j=1}^{N}\mu_{ij}\boldsymbol{x}_j \Big/ \sum_{j=1}^{N}\mu_{ij} \tag{3}$$

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New York: ACM,
Boston: Addison-Wesley Professional,
Washington, DC: AMIA Publications,
Arlington: AUAI Press,
Berlin: Springer, Cham: Springer
Nottingham, UK: BMVA Press,
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Cambridge, MA: MIT Press,
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London: dblp Computer Science Bibliography,
San Jose, CA: EDA Consortium,
New York: Elsevier Science Inc.,
Singapore: IACSIT Press,
New York: IEEE Communications Society,
Washington, DC: IEEE Computer Society,
Piscataway, NJ: IEEE,
Amsterdam: IOS Press,
Hoboken, NJ: John Wiley & Sons,
Hingham, MA: Kluwer Academic Publishers,
New York: McGraw-Hill Higher Education,
San Francisco, CA: Morgan Kaufmann Publishers Inc.,
New York: Oxford University Press,
Upper Saddle River, NJ: Prentice-Hall,
Princeton, NJ: Princeton University Press,
London: Routledge,
Philadelphia, PA: SIAM,
Bellingham, WA: SPIE,
Berkeley, CA: USENIX Association,
River Edge, NJ: World Scientific Publishing Co., Inc.,

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(SVM)
Support Vector Machine (SVM)
(Support Vector Machine, SVM)

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A 50% B 75%

a) “ A B 25 ”
75%-50%=25% 25

b) “ A B 50%”
(75%-50%)/50%=25%/50%=50% 50%

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(ICIR-UMBCEEA) (DMRS) (CoMP) CoMP (IRC) IRC CoMP (UMBCA-OWFC) ICIR-UMBCEEA 19.2% 24.5% ICIR-UMBCEEA 25.48% 18.03% ICIR-UMBCEEA UMBCA-OWFC ICIR-UMBCEEA

Since the energy consumption of joint processing in uplink multi-base cooperative communication system is excessively high, an Inter-Cell Interference Rejection based Uplink Multi-Base Cooperative Energy Efficiency Algorithm (ICIR-UMBCEEA) was proposed. Firstly, equivalent noise and Coordinated Multi-Point (CoMP) estimated channel were gotten by DeModulation Reference Signal (DMRS) sequence, and the Interference Rejecting Combining (IRC) filtering matrix of CoMP channel was deduced; Secondly, an equivalent interference model was established and the average inter-cell interference was obtained by using IRC filtering matrix; Finally, interference level of user in each cell to non-CoMP set was calculated, and a joint processing against strong interference users was made. In the comparison experiments with Uplink Multi-Base Cooperative Algorithm of Optimal Water Filling Control (UMBCA-OWFC), the normalized average interference of ICIR-UMBCEEA decreased by 19.2% in center users and 24.5% in edge users, and the energy efficiency of it increased by 25.48% in center users and 18.03% in edge users; ICIR-UMBCEEA had less energy consumption, and had higher throughput in center users and not much difference in edge users. The theoretical analysis and simulation results show that ICIR-UMBCEEA can effectively improve the energy efficiency of communication system in engineering.

(Pareto)

(FMODE) ZDTl ZDT4 ZDT6 (500) FMODE NSGA Ⅱ FMODE NSGA Ⅱ SPEA Ⅱ DEMO FMODE FMODE FMODE

Concerning the high time-complexity of multi-objective evolutionary algorithm based on Pareto dominated solution sorting, considering the potential features of non-dominated solution sorting, a fast sorting method which only handles individuals with the highest rank in current population was introduced. The individuals could be chosen into the next generation during the sorting. The algorithm was terminated when the population of next generation was selected enough, which reduced the number of individuals for sorting process and the time complexity. In addition, a method of uniform crowding distance calculation was given. Finally, a Fast Multi-Objective Differential Evolution (FMODE) algorithm was proposed which incorporated the introduced sorting method and uniform crowding distance into Differential Evolution (DE). Simulation experiments were conducted on the standard multi-objective optimization problems including ZDTl~ZDT4 and ZDT6. The time consumption of FMODE was far less than NSGA Ⅱ in large size of population, and its overall performance was much better than the classic NSGA Ⅱ, SPEA Ⅱ and DEMO. Moreover, the uniform crowding distance method provided better performance than classic crowding distance in framework of FMODE. The parameters of FMODE were also obtained by experiments. The simulation results show that the proposed algorithm has lower time complexity of calculating level, and better performance in convergence and diversity.

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1 Li Jiangning
2 Einstein A
3 Williams-Ellis A
4 De Morgan A
5 Godard J-L
6 Peebles P Z Jr.

P32

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B

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